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Type(Objective)

1 Which of the following is incorrect for frogs? 887527

The vascular system of frog is well-developed, closed type

The blood vascular system involves heart, blood vessels and blood

The lymphatic system consists of lymph, lymph channels and lymph nodes

The lymph is different from blood and has just few proteins and RBCs

2 Match the columns: 887526

Column – I		Column – II	
(A)	Fore-brain	(i)	A pair of optic lobes
(B)	Mid-brain	(ii)	Cerebellum and medulla oblongata
(C)	Hind-brain	(iii)	Olfactory lobe, paired cerebral diencephalon

A - (ii), B - (iii), C - (i)

A - (ii), B - (i), C - (iii)

A - (iii), B - (i), C - (ii)

A - (i), B - (ii), C - (iii)

3 The most common species of frog found in India is: 887525

Rana cyanophlictis

Rana pipiens

Rana sylvatica

Rana tigrina

4 Which of the following is correct for the digestion in the alimentary canal of frog? 887524

Digestion of food takes place by the action of HCl and gastric juices secreted from the walls of the- stomach

Partially digested food called chyme

The duodenum receives bile and pancreatic juices through a common bile duct

More than one option is correct

5 Which of the following statements is/are correct for respiration in frogs? 887523

On land, the buccal cavity, skin and lungs act as the respiratory organs

The lungs are a pair of pink-coloured structures present in the upper part of thorax

Both A and B

Pulmonary respiration is absent in frogs

6 The frog: 887522

Excretes urea

A pair of kidneys is located posteriorly, one kidney on either side of vertebral column

Each kidney is composed of several structural and functional units called uriniferous tubules or nephrons

Possess all of the above features

7 Which of the following is true for the excretory system of frogs? 887521

Two ureters emerge from the kidneys of male frogs which acts as urinogenital duct that opens into cloaca

In female frogs, ureters and oviduct form the urinogenital duct that opens into cloaca

Both A and B

Urinary bladder is present dorsal to the rectum which open into the cloaca

8 Choose the correct statement for female reproductive system of frog: 887520

Ovaries have no functional or internal connections with the kidneys

Possess Bidder's canal

The posterior ends of oviduct are broad called ovisacs which open dorsally into cloaca

More than one option is correct

9 In frogs: 887519

Respiration by lungs is called pulmonary respiration

The lungs are a pair of elongated, pink-coloured sacs present below the thorax

During aestivation and hibernation gaseous exchange takes place through skin

Both A and C

10 Which of the following is incorrect for the vascular system of frogs? 887518

Arteries carry blood away from the heart to all body parts

Conus arteriosus or truncus arteriosus receives blood from ventricle

Sinus venosus joins the left atrium

All of the above

11 The excretory system of frog consists of: A pair of kidneys 887517

A pair of kidneys

Ureters

Urinary bladder and cloaca

All of the above

12 Choose the correct statement for the blood circulation in frog's heart: 887516

The ventricle opens into a sac-like conus arteriosus on the ventral side of the heart

A triangular structure called sinus venosus joins the right atrium

Sinus venosus receives blood from vena cava

More than one option is correct

13 Read the following statements with respect to venous system of frogs: 887515

(i) Heart is a muscular structure situated in the upper part of the body cavity.

(ii) Heart has four chambers.

(iii) Heart is covered by a membrane called pericardium.

(iv) Renal portal vein carry blood from lower parts of the body.

(v) Special venous connections between liver and intestine as well as kidney and lower parts are present.

How many of the above statements are correct?

1

4

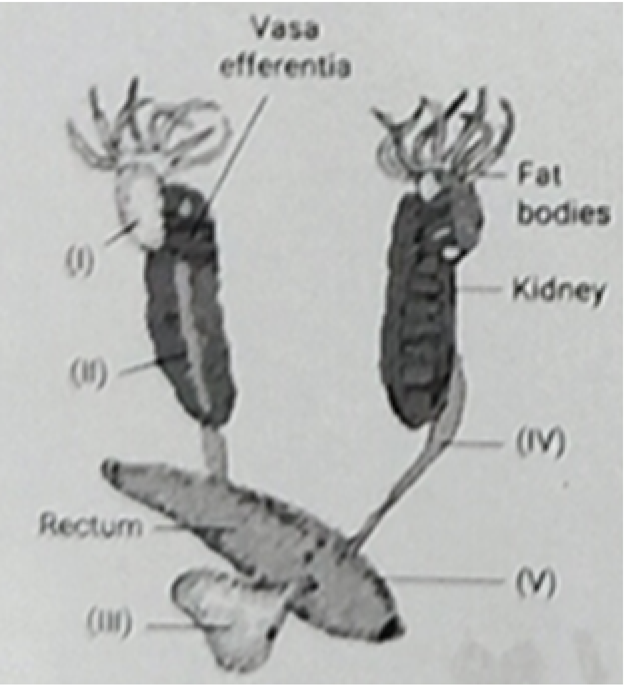
5

3

14 **Which of the following is correct for CNS of frogs?** 887514

- Medulla oblongata passes out through the foramen magnum and continues as spinal cord
- Mid brain possesses one optic lobe
- Brain contains 10 cranial nerves
- All are correct

15 **Identify the structures labelled as I, II, III, IV and V for frogs:** 887513



- I - Testes, II - Cloaca, III - Adreanal gland, IV - Urinary bladder, V - Urinogenital duct
- I - Testes, II - Cloaca, III - Urinary bladder, IV - Adreanal gland, V - Urinogenital duct
- I - Testes, II - Adreanal gland, III - Urinary bladder, IV - Urinogenital duct, V - Cloaca
- I - Testes, II - Urinary bladder, III - Adreanal gland, IV - Cloaca, V - Urinogenital duct

16 Which of the following is correct for frog? 887512

The hind limbs are larger and muscular than fore limbs

Hind limbs are shorter than fore limbs

Hind limbs bear five digits with web

Both A and C

17 Which of the following is correct for frogs? 887511

Has ureter to carry both urine and sperm

Bidder's canal is present inside the kidneys which carry sperms

Both A and B

These possess monocondylic skull

18 In frogs: 887510

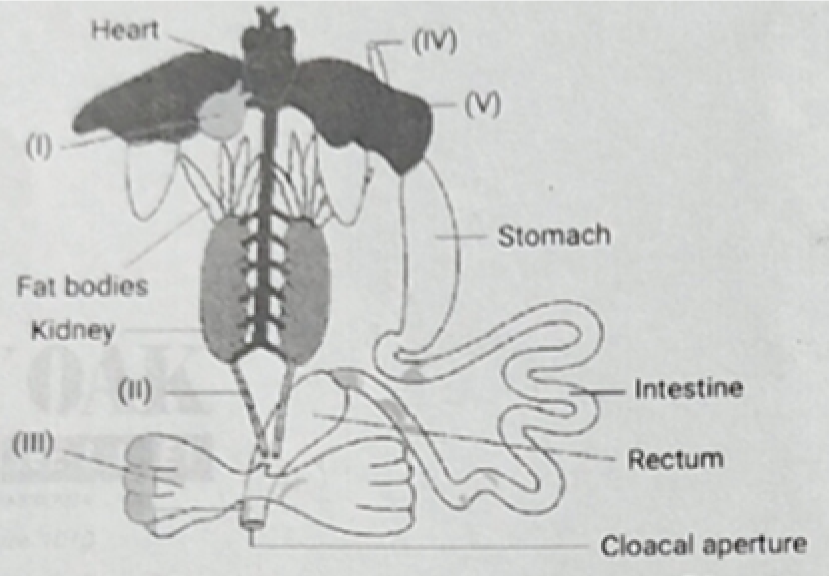
Skin acts as respiratory organ in water

Dissolved O_2 in the water is exchanged through the skin by active transport

Both A and B are correct

Cutaneous respiration is absent

19 **Identify the structures I, II, III, IV and V for the diagrammatic representation of internal organs of frog:** 887509



I - Gall bladder, II - Ureter, III - Cloaca, IV - Oesophagus, V - Liver

I - Gall bladder, II - Cloaca, III - Ureter, IV - Mouth, V - Crop

I - Structure for storing bile, II - Ureter, III - Urinary bladder, IV - Oesophatus, V - Liver

I - Urinary bladder, II - Ureter, III - Structure for storing bile, IV - Oesophatus, V - Gizzard

20 **The alimentary canal in frog is:** 887508

Complete, i.e., from mouth to cloaca

Short because frogs are carnivores

Both A and B

Incomplete

21 Choose the correct statement for frog: 887507

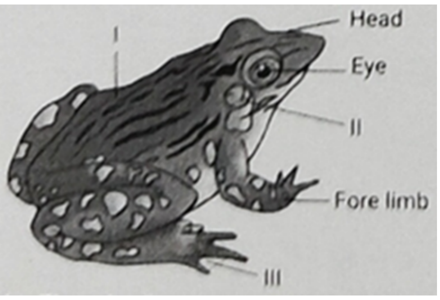
Body of a mature frog is divisible into head, thorax and abdomen

On the either side of eyes a membranous tympanum is present which receives the sound signals

The fore limbs have no role in swimming

The hind limbs end in four digits

22 Identify I, II and III respectively: 887506



Trunk, Tympanum, Web

Neck, Brown eye spot, Web

Trunk, Tympanum, Hind limb

Neck, Tympanum, Hind limb

23 The frog's heart is enclosed within a membrane called: 887505

Sinus venosus

Pleura

Mesorchium

Pericardium

24 Read the following statements (assertion-A and reason-R) and select the correct option. 887504

A: The digits of forelimbs of a frog possess web.

R: The webs found in the digits of forelimb help in swimming.

Both A and R are true and R is correct explanation of A

Both A and R are true but R is not the correct explanation of A

A is true but R is false

Both A and R are false statements

25 Renal portal vein in frogs: 887503

Carry oxygenated blood

Carry blood from lower part of the body to kidneys

Both A and B are correct

May bypass kidney and directly enter heart

26 The respiratory organs of frog is/are: 887502

Skin

Lungs

Buccal cavity

All of the above

27 The skin of frog: Is smooth and slippery due to the presence of mucus 887501

Is smooth and slippery due to the presence of mucus

In the dorsal side is usually pale yellow

In the ventral side is usually olive green with dark irregular spots

Is related to all of the above statements

28 Which of the following is correct for frog? 887500

Frogs are beneficial for mankind because they eat insects and protect the crop

Frogs maintain ecological balance because these serve as an important link of food chain and food web in the ecosystem

In some countries, the muscular legs of frog are used as food by man

All of the above

29 **The undigested solid waste in frogs:** 887499

From ileum moves into duodenum

Passes out through cloaca

Passes out through well-developed anus

From rectum passes out through urinary bladder

30 **Testes in frogs: Are attached to the kidney directly through their external surface** 887498

Are attached to the kidney directly through their external surface

Are attached to the kidney to the upper part by mesorchium

Both (A) and (B)

Are triangular shaped

31 **Read the following statements for frogs:** 887497

(i) A mature female can lay 2500 to 3000 ova at a time.

(ii) Fertilization is external and occurs in water.

(iii) Development is indirect.

(iv) Development involves a larval stage called tadpole.

(v) Tadpole undergoes metamorphosis to form the adult.

How many of the above statements are correct?

2

4

3

5

32 Male frogs can be distinguished from female frogs by the following characters: 887496

Males possess sound producing vocal sacs to produce characteristic sound called croacking

Males possess copulatory pads on fore limbs

Copulatory pads on fore limbs are absent in female frogs

More than one option is correct

33 Frog is amphibian and has copious amounts of water at its disposal for elimination of nitrogenous waste. It is: 887495

Uricotelic

Ammonotelic

Ureotelic

Guanotelic

34 Read the following statements (assertion-A and reason-R) and select the correct option. 887494

A: The alimentary canal of frog is short.

R: Frog is carnivorous. Both A and R are true and R is correct explanation of A

- Both A and R are true but R is correct explanation of A
- Both A and R are true but R is not the correct explanation of A
- A is true but R is false
- Both A and R are false statements

35 Choose the correct statement for the digestive system of frog: 887493

- Oesophagus is very long
- Food is captured by the bilobed tongue
- Both A and B
- Final digestion takes place in rectum

36 Eyes of frogs: 887492

- Are bulged and lack nictitating membrane
- Possess nictitating membrane that protects them while in water
- Provide them with 180□ vision
- Are compound eyes

37 **In frogs:** 887491

Digestion of food takes place by action of HCl and gastric juice in stomach

Partially digested food called chyme is passed from stomach to the first part of intestine, the ileum

Digested food is absorbed by the numerous finger like folds in the inner wall of intestine called vi and microvilli

Both A and C are correct

38 **Frogs:** 887490

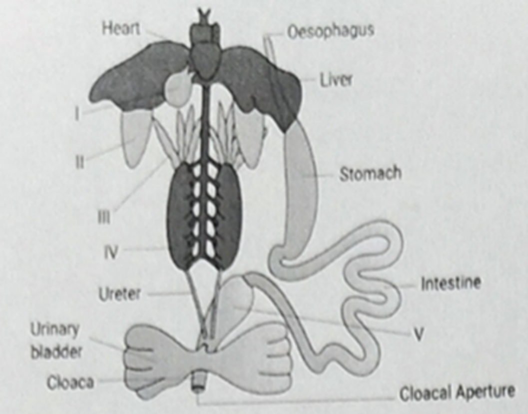
Exhibit sexual dimorphism

Have hind limbs and fore limbs which help in walking, leaping and burrowing

Have webbed digits that help in swimming

Show all of the above features

39 **The above figure is associated with diagrammatic representation of internal organs of frog. Identify I to V.** 887489



I - Gall bladder, II - Lung, III - Ovary, IV - Testis, V - Rectum

I - Gall bladder, II - Lung, III - Fat bodies, IV - Testis, V - Rectum

I - Gall bladder, II - Lung, III - Testis, IV - Kidney, V - Rectum

I - Gall bladder, II - Lung, III - Fat bodies, IV - Kidney, V - Rectum

40 Read the following statements for frog and answer accordingly: 887488

I. Duodenum receives bile and pancreatic juice through hepatopancreatic duct.

II. Bile emulsifies fat and pancreatic juices digest carbohydrates and proteins.

Both the statements I and II are correct and II is the correct explanation of I

Both the statements I and II are correct but II is not the correct explanation of I

Statement I is correct but II is incorrect

Both the statements I and II are incorrect

41 Read the following statements for frog: 887486

(i) They possess both neural and endocrine system for control and coordination.

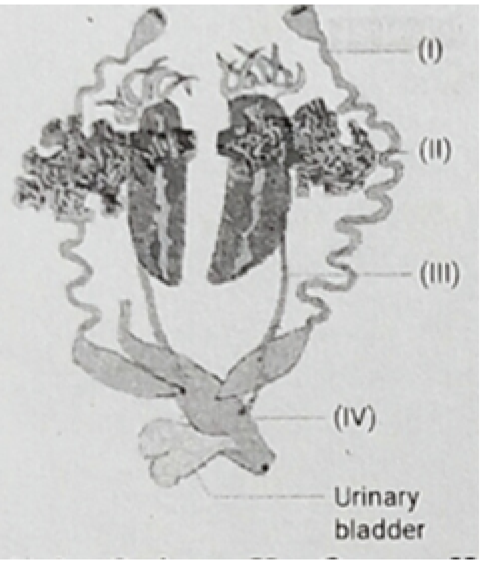
(ii) They possess a central nervous system (Brain and spinal cord) and a peripheral nervous system
(carnial and spinal nerves).
(iii) They lack an autonomic nervous system.
(iv) They possess 10 pairs of cranial nerves arising from the brain.
(v) The hindbrain of frog consists of a pair of optic lobes.
How many of the above statements are correct?

- Two
- Three
- Four
- Five

42 Select the correct statement for frogs: 887485

- Ear acts as organ for hearing but not balancing
- External ear is absent and only tympanum can be seen externally
- They possess compound eyes
- All of the above

43 Identify I, II, III and IV for frogs: 887484



I - Ovisac, II - Ovary, III - Cloaca, IV - Ureter

I - Oviduct, II - Ovary, III - Urinogenital tract, IV - Cloaca

I - Ovisac, II - Cloaca, III - Oviduct, IV - Anus

I - Oviduct, II - Ovary, III - Ureter, IV - Cloaca

44 Vasa efferentia in frogs: 887483

Are 30-40 in number connecting testes to the kidneys

Run transversely through the mesorchium and opens into Bidder's canal

Do not connect to Bidder's canal

Has no role in sperm transport

45 The heart of frog: 887482

Has four chambers - two atria and two ventricles

Has two chambers - one atrium and one ventricle

Has two atria and one ventricle

Has one atrium and two ventricles

46 Statement I: Fungi are generally aquatic 887481
Statement II: Zoospores is a sexual spore of Kingdom Fungi.

Both statements are correct.

Both statements are incorrect.

Statement I is correct but Statement II is incorrect.

Statement I is incorrect but Statement II is correct.

47 Which protozoan possesses an infectious spore-like stage? 887480

Trypanosoma

Plasmodium

Paramoecium

amoeba

48 Which one of the following statements is incorrect regarding fungi? 887479

They are heterotrophic organisms

All fungi are multicellular

They occur in air, water and soil

They prefer warm and humid places.

49 Statement 1: three-domain system divides the Kingdom Monera into three domains. 887478

Statement 2: bacteria and blue green algae were under kingdom animalia in earlier classification system.

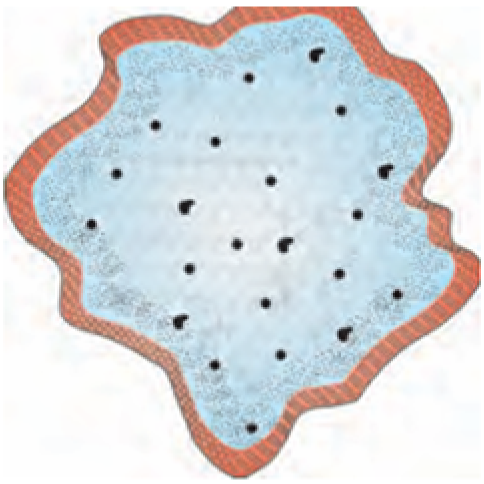
Both Statements are correct

Both Statements are incorrect.

Statement 1 is incorrect but Statement 2 is correct

Statement 1 is correct but Statement 2 is incorrect

50 Pitcher belongs to which group of kingdom Protista. 887477



Slime Mould

Euglenoids

Dinoflagellates

None of the above

51 Match the following columns. **887476**

Column-1	Column-2
A. Slime mould	1. red tide
B. Gonyaulax	2. Saprophytic
C. Amoeba	3. cilia
D. Paramoecium	4. pseudopodia

A-1, B-2, C-3, D-4

A-2, B-3, C-4, D-1

A-2, B-1, C-4, D-3

A-3, B-4, C-2, D-1

52 Match the following columns. **887475**

Column-1	Column-2
A Halophiles	1. Hot springs
B.Thermoacidophiles	2. salty areas
C. Methanogens	3.Biogas production

A-1, B-2, C-3

A-1, B-3, C-2

A-2, B-1, C-3

A-3, B-2, C-1

53 **Assertion (A): Diatomaceous earth is useful in polishing and filtration.**
Reason (R): Diatom walls contain indestructible silica.

887474

Both A and R are true and the Reason is the correct explanation of the Assertion.

Both A and R are true but the Reason is not the correct explanation of the Assertion.

Assertion is true but Reason is false.

Both Assertion and Reason are false.

54 **Out of the following how many characters were chosen by R.H Whittaker for his 5 kingdom classification-**

887473

1. presence or absence of cilia
2. cell structure
3. Mode of nutrition
4. Reproduction
5. habitat

2

1

4

3

55 **Spores of slime moulds are highly resistant because they possess:** 887472

Silica wall

True wall

Chitin wall

Polysaccharide wall

56 **Statement 1:** ***Paramoecium*** **is aquatic and motile organism.** 887471

Statement 2: Penicillium is a antibiotic that we obtain from fungus.

Both Statements are correct

Both Statements are incorrect.

Statement 1 is incorrect but Statement 2 is correct

Statement 1 is correct but Statement 2 is incorrect

57 **Statement 1: sexual reproduction in protista take place by a process involving cell fission and zygote Formation** 887470

Statement 2: All single-celled eukaryotes are placed under phylum Protista.

Both Statements are correct

Statement 1 is correct but Statement 2 is incorrect

Statement 1 is incorrect but Statement 2 is correct

Both Statements are incorrect.

58 Which of the following are not included in any of the kingdom? 887469

Bacteria

Viruses

Slime moulds

Protozoans

59 Marine amoeboid protozoans possess: 887468

Cell wall embedded with silica

Silica shell

Protein rich pellicle

Chitinous cell wall

60 How many of these are not bacterial disease Cholera, malaria, typhoid, tetanus, citrus canker ? 887467

3

4

1

2

61 Which statement about slime moulds is incorrect? 887466

Fruiting bodies develop under favourable conditions

They are saprophytic

Spores disperse by air currents

Plasmodium may spread several feet

62 How many of the following are photosynthetic? Diatoms, Dinoflagellates, Slime moulds, Amoeba 887465

1

3

2

4

63 Statement 1: Fungi have non cellulosic cell wall. 887464

Statement 2: Kingdom Fungi and Kingdom Animalia have only heterotrophic mode nutrition.

Both Statements are incorrect

Statement 1 is correct but Statement 2 is incorrect

Statement 1 is incorrect but Statement 2 is correct

Both Statements are correct.

64 Which of these does not form basis for the division of the kingdom into various classes? 887463

morphology of the mycelium

mode of nutrition

fruiting bodies

None of the above

65 Which one possesses pigments identical to higher plants? 887462

Chrysophytes

Dinoflagellates

Euglenoids

Slime moulds

66 How many Statements are correct about sexual reproduction in fungi. 887461

A. Gametes undergoing plasmogamy are haploid

B. Dikaryophase is found in Ascomycetes

C. Fusion of two nuclei is called karyogamy

D. Spores are haploid

2

1

3

4

67 How many of these organisms are Unicellular? 887460

Lichen, Desmids, Yeast, Chlamydomonas, Euglena, Mycoplasma, Spirogyra, Mycorrhiza

5

4

7

6

68 Which of the following is incorrect about cyanobacteria? 887459

They donot form colonies

They can fix atmospheric nitrogen

They can do photosynthesis

They form bloom in polluted water body

69 Which group of organisms have a cavity called Gullet? 887458

Sporozoans

Ciliated protozoans

Flagellated protozoans

Amoeboid protozoans

70 Statement 1: Nostoc can do nitrogen fixation but not photosynthesis. 887457
Statement 2: Mycorrhiza is a symbiotic association.

Both Statements are incorrect

Statement 2 is correct but Statement 1 is incorrect

Statement 2 is incorrect but Statement 1 is correct

Both Statements are correct.

71 how many of these feature can be basis for separation of Kingdom Fungi and Kingdom Plantae? 887456

Eukaryotic Cell

Mode of Nutrition

Presence of Cell wall

Cellular level of organization

1

2

3

4

72 Red tides are produced due to: 887455

Dinoflagellates

Diatoms

Euglenoids

Slime moulds.

73 which type of bacteria is represented by given pitcher? 887454

Baccilus

Coccus

Spirillum

Vibrium

74 Dikaryophase in some fungi is represented by- 887453

2n

n + n

2n +2n

2n + n

75 Pellicle in Euglena is composed mainly of – 887452

Polysaccharides

Chitin

Protein

Cellulose

76 An infectious spore like stage is present in the life cycle of a protozoan which causes- 887451

Sleeping sickness

cholera

Malaria

typhoid

77 Sleeping sickness is caused by: 887450

Amoeba

Entamoeba

Trypanosoma

Plasmodium

78 In fungi asexual reproduction take place – 887449

conidia, sporangiospores and oospore

ascospore, sporangiospores and oospore

conidia, zoospore, oospore

conidia, sporangiospores and zoospore

79 Which of the following group belongs to protozoans? 887448

euglenoids, flagellates, ciliates, sporozoans eubacteria, ciliates, desmids, cynobacteria

archaebacteria, flagellates, ciliates, dinoflagellates

amoeboid, flagellates, ciliates, sporozoans

80 Statement 1: In absence of sunlight euglena show parasitic mode of nutrition
Statement 2: Parasitic forms can be found in amoeboid protozoans.

887447

Both Statements are correct

Statement 2 is correct but Statement 1 is incorrect

Statement 2 is incorrect but Statement 1 is correct

Both Statements are incorrect.

81 Most of the bacteria obtain nutrition from:

887446

Sunlight

inorganic substrates

dead or decaying matter

plants

82 **How many of the following possess a cell wall? Mycoplasma, Paramoecium, Desmids, Cynobacteria, Gonyaulax, Amoeba** 887445

2

3

4

5

83 **Statement 1: Both Euglena and cyanobacteria are photosynthetic**
Statement 2: All members of flagellated protozoans are Parasitic. 887444

Both Statements are correct

Statement 2 is correct but Statement 1 is incorrect

Statement 2 is incorrect but Statement 1 is correct

Both Statements are incorrect.

84 **Assertion (A): Both cyanobacteria and Euglena perform photosynthesis**
Reason (R): Both possess chlorophyll a. 887443

Both A and R are true and the Reason is the correct explanation of the Assertion.

Both A and R are true but the Reason is not the correct explanation of the Assertion.

Assertion is true but Reason is false.

Both Assertion and Reason are false.

85 **Which characteristic is common to both bacteria and protists?** 887442

Membrane bound nucleus

Can produce bio gas

Cell membrane

Tissue organization

86 **Cell wall of Fungi is made up of –** 887441

Only Chitin

Only Cellulose

Chitin and Cellulose

Chitin and Polysaccharide

87 **Methanogens naturally occur in** 887440

acidic lakes

human intestine

Rumen of cattle and buffalo

volcanic soil

88 The organisms placed under Monera lacks 887439

Ribosome

Genetic material

Membrane bound nucleus

Membrane less nucleus

89 The major reason fungi were separated from Plantae is – 887438

absence of well defined nucleus

heterotrophic nutrition and chitinous cell wall

absence of membrane bound organelles

lack of reproduction

90 Statement 1: Euglenoids have cellulose cell walls.
Statement 2: Diatoms are chief Consumer of oceans. 887437

Both Statements are correct

Statement 1 is correct but Statement 2 is incorrect

Statement 1 is incorrect but Statement 2 is correct

Both Statements are incorrect.

91 Dipole moment is highest for: 887436

$CHCl_3$

CH_4

CHF_3

CCl_4

92 H-bonding is not present in: 887435

NH_3

Water

H_2S

HF

93 The bond strength in O_2^+, O_2, O_2^- and O_2^{2-} follows the order: 887434

$O_2^{2-} > O_2^- > O_2 > O_2^+$

$O_2^+ > O_2 > O_2^- > O_2^{2-}$

$O_2 > O_2^- > O_2^{2-} > O_2^+$

$O_2^- > O_2^{2-} > O_2^+ > O_2$

94 Which is not a paramagnetic species? 887433

O_2

O_2^+

O_2^-

O_2^{2-}

95 The ratio of σ and π-bonds in benzene is: 887432

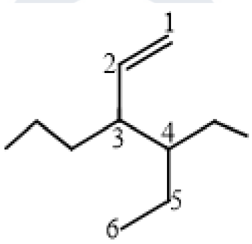
2

6

4

8

96 The correct IUPAC name of the compound is: 887431



is :

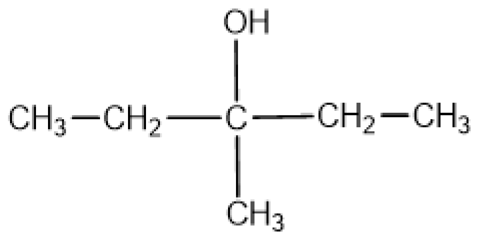
3-(1-Ethylpropyl)hex-1-ene

4-Ethyl-3-propylhex-1-ene

3-Ethyl-4-ethenylheptane

3-Ethyl-4-propylhex-5-ene

97 Write the IUPAC name of the compound: 887430



3-Methylpentane-3-ol

3-Hydroxyhexane

3-Hydroxy-3-methylpentane

All of the above

98 The IUPAC name of CH_3–$CH(CH_2CH_3)$–CHO is: 887429

Butan-2-aldehyde

2-Methylbutanal

3-Methyl isobutyraldehyde

2-Ethylpropanal

99 The IUPAC name of $CH_3\_CH_2–CH(CH_3)–NH_2$ is: 887428

1-Methyl-1-aminopropane

Butan-2-amine

2-Methyl-3-aminopropane

None of the above

100 The number and type of carbon atoms present in neopentane are: 887427

Four 1° carbons, one 4° carbon

Two 1° carbons, two 2° carbons

One 1° carbon, one 4° carbon

One 1° carbon, one 4° carbon

101 Functional group present in amides is: 887426

$–COOH$

$–NH_2$

$–CONH_2$

$–COO^-$

102 **Which is a saturated compound?** 887425

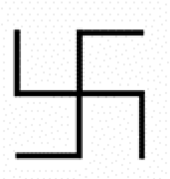
Alkanes

Alkenes

Alkynes

Cycloalkenes

103 **The IUPAC name for the hydrocarbon represented by the swastik sign is:** 887424



Neo-nonane

Tetraethylcarbon

2-Ethylpentane

3,3-Diethylpentane

104 **The suffixes for alcohols, aldehydes and ketones, according to the IUPAC system are respectively:** 887423

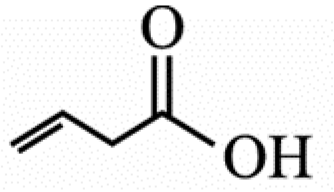
-ane, -al, -keto

-ol, -al, -keto

-al, -ol, -one

-ol, -al, -one

105 **The IUPAC name of the compound is:** 887422



But-3-enoic acid

But-1-enoic acid

Pent-4-enoic acid

Prop-2-enoic acid

106 **The IUPAC name of CH_3COCH_3 is:** 887421

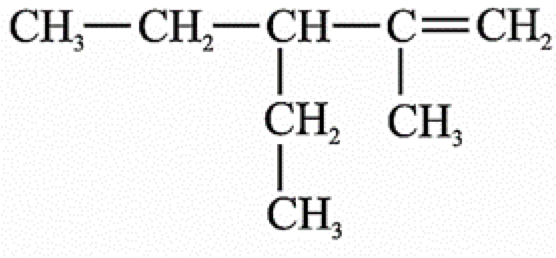
Acetone

2-Propanone

Dimethyl ketone

Propanal

107 **The IUPAC name is:** 887420



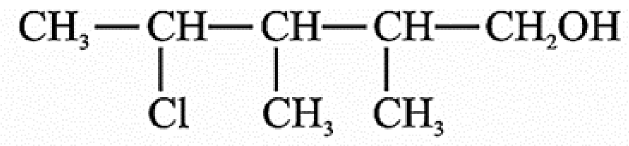
2-Methyl-3-ethylpent-1-ene

3-Ethyl-4-methylpent-4-ene

3-Ethyl-2-methylpent-1-ene

3-Methyl-2-ethylpent-1-ene

108 **The correct IUPAC name of the organic compound is:** 887419



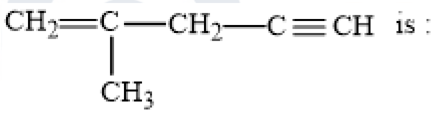
4-Chloro-2,3-dimethylpentan-1-ol

2-Chloro-3,4-dimethylpentan-5-ol

2,3-Dimethyl-4-chloropentan-1-ol

2-Chloro-3,4-dimethyl n-pentyl alcohol

109 **The IUPAC name of the compound is:** 887418



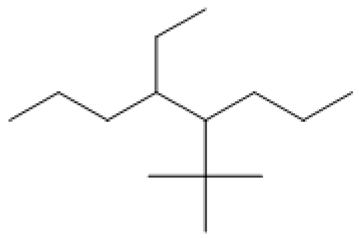
2-Methylpent-1-en-4-yne

4-Methylpent-4-en-1-yne

2-Methylpent-2-en-4-yne

4-Methylpent-1-en-4-yne

110 Correct IUPAC name of the following compound is: 887417



4-(1,1-Dimethylethyl)-5-ethyloctane

5-Ethyl-4-(1,1-dimethylethyl)octane

5-(1,1-Dimethylethyl)-4-ethyloctane

5-Ethyl-5-(1,1-dimethylethyl)octane

111 Acetylene molecules contain: 887416

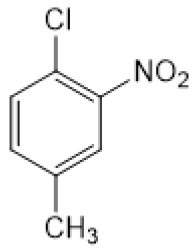
5σ bond

4σ bond and 1π bond

3σ and 2π

3σ and 3π

112 **The IUPAC name for the compound is:** **887415**



1-Chloro-2-nitro-4-methylbenzene

1-Chloro-4-methyl-2-nitrobenzene

2-Chloro-1-nitro-5-methylbenzene

m-Nitro-p-chlorotoluene

113 **The IUPAC name of the compound $CH_3–CH=CH–C≡CH$ is:** **887414**

Pent-3-en-1-yne

Pent-2-en-4-yne

Pent-1-yn-3-ene

Pent-4-yn-2-ene

114 **The number of olefinic bonds in the given compound is/are:** **887413**

$$H_2C = CH - \underset{\underset{O}{\|}}{C} - CH = CH - CH = NH$$

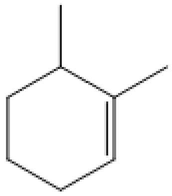
2

3

1

4

115 The IUPAC name of the structure is: **887412**



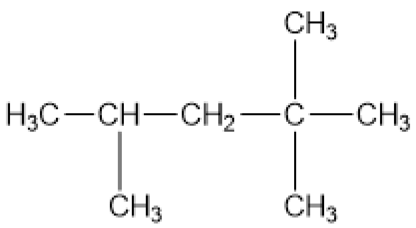
1,2-Dimethylcyclohexane

1,6-Dimethylcyclohexene

1,2-Dimethylcyclohex-2-ene

2,3-Dimethylcyclohexane

116 How many primary, secondary and tertiary carbon atoms are present in the following compound? **887411**



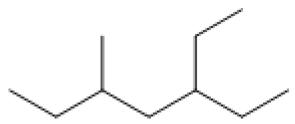
One primary, two secondary and one tertiary

Five primary, three secondary

Five primary, one secondary, one tertiary and one quaternary

Four primary, two secondary and two quaternary

117 The correct IUPAC name of the following compound is: 887410



3-Ethyl-5-methylheptane

2,4-Diethylhexane

3-Methyl-5-ethylheptane

3,5-Diethylhexane

118 The IUPAC name of the compound is: 887409

$$\underset{\displaystyle OH}{\underset{|}{CH_2}} - \underset{\displaystyle OH}{\underset{|}{CH}} - \underset{\displaystyle OH}{\underset{|}{CH_2}}$$

1,2,3-Trihydroxypropane

3-Hydroxypentane-1,5-diol

1,2,3-Hydroxypropane

Propane-1,2,3-triol

119 Which one is not correct for a homologous series? 887408

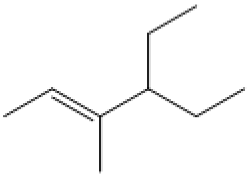
All members have same general formula

All members have same chemical properties

All members have same physical properties

All members have same functional group

120 The IUPAC name of the following compound is: **887407**



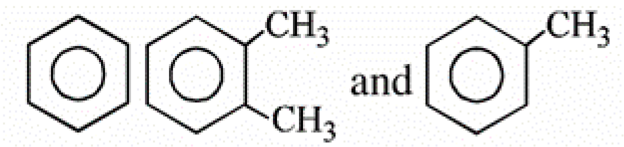
4-Methyl-3-ethylhex-4-ene

4,4-Diethyl-3-methylbut-2-ene

3-Ethyl-4-methylhex-4-ene

4-Ethyl-3-methylhex-2-ene

121 Number of secondary carbon atoms present in the above compounds are respectively: **887406**



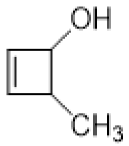
6, 4, 5

4, 5, 6

5, 4, 6

6, 2, 1

122 The IUPAC name of the compound is: 887405



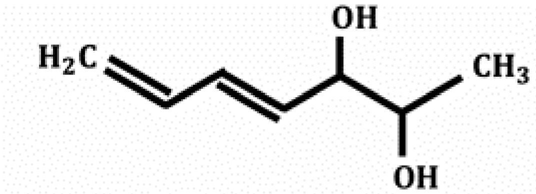
3-Methyl cyclobutene-2-ol

4-Methyl cyclobut-2-en-1-ol

4-Methyl cyclobuten-3-ol

2-Methyl cyclobut-3-en-1-ol

123 What is the IUPAC name of the molecule shown? 887404

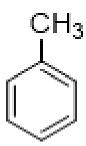


1,3-Heptadiene-5,6-diolU

4,5-Heptadiene-2,3-diol 4-Heptene-2,3-diol

4-Heptene-2,3-diol

4,6-Heptadien-2,3-diol

124 How many 1°, 2° & 3° H atoms are present in  respectively? 887403

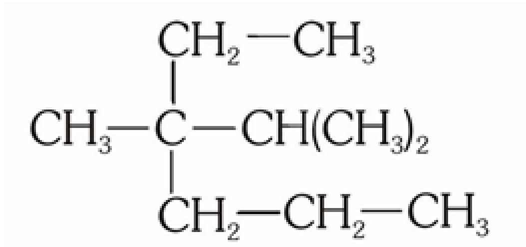
3, 0, 5

3, 5, 0

4, 3, 0

0, 5, 3

125 **The IUPAC name of the given structure is:** **887402**



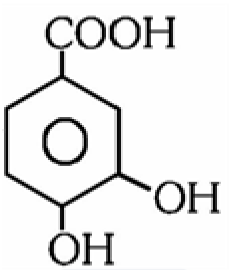
3-Methyl-3-isopropylhexane

3-Isopropyl-3-methylhexane

3-Ethyl-2,3-dimethylhexane

2,3-Dimethyl-3-ethylhexane

126 **The IUPAC name of the compound is:** **887401**



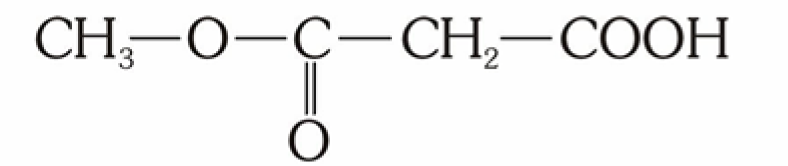
3,4-Dihydroxy benzoic acid

2,3-Dihydroxy benzoic acid

4-Carboxy-2-hydroxy phenol

4-Carboxy benzene-1,2-diol

127 The IUPAC name of the above compound is: 887400



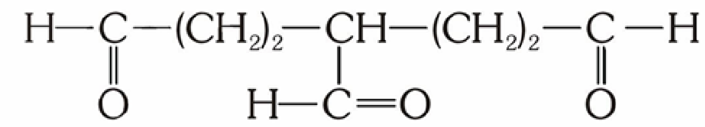
2-Acetoxy ethanoic acid

2-Methoxycarbonyl ethanoic acid

3-Methoxyformyl ethanoic acid

2-Methoxyformyl acetic acid

128 The IUPAC name of the compound is: 887399



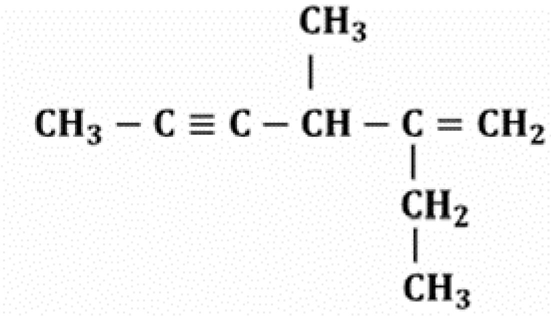
1,4,7-Triheptanal

1,3,5-Triformyl pentane

3-Formyl heptane dial

1,3,5-Pentane Tri carbaldehyde

129 The IUPAC name of the given compound is: 887398



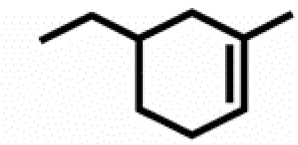
2-Ethyl-3-methyl-hexa-1-en-4-yne

5-Ethyl-4-methyl-hexa-2-yn-5-ene

3-Methyl-4-methyl-hexa-2-yn-5-ene

4-Methyl-5-methylene hepta-2-yne

130 Correct IUPAC name is: 887397



1-Methyl-3-ethylcyclohexene

5-Ethyl-1-methylcyclohexene

2-Ethyl-4-methylcyclohexene

3-Ethyl-1-methylcyclohexene

131 Two masses of 1 gm and 4 gm are moving with equal kinetic energies. The ratio of the magnitudes of their linear momenta is 887396

4 : 1

$\sqrt{2}:1$

1 : 2

1 : 16

132 The potential energy of a particle is given by $U(x)=5x^2 - 4x+3$ J. The force acting on the particle at x=2 m is: 887395

-16 N

-20 N

16 N

20 N

133 Mechanical energy of a system remains constant only when: 887394

Only conservative forces do work.

Only non-conservative forces do work.

The net force is zero.

The acceleration is zero.

134 A 5 kg block slides down a frictionless incline from a vertical height of 1.8 m. Taking $g = 10\ m/s^2$, its speed at the bottom is: 887393

4 m/s

6 m/s

8 m/s

10 m/s

135 **A 1 kg block attached to a spring of spring constant 100 N/m is stretched by 20 cm and released. The kinetic energy of the block as it passes through the mean position is:** 887392

1 J

2 J

3 J

4 J

136 **A 5 g particle moves according to $x = 3t + 4t^2 - t^3$. The work done by the net force from t=0 to t=2 s is:** 887391

0.08 J

0.10 J

0.12 J

0.14 J

137 The value of 5 J in erg is: 887390

5×10^5 erg

5×10^6 erg

5×10^7 erg

5×10^8 erg

138 A force vector applied on a mass is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ and accelerates with $1 m/s^2$. What will be the mass of the body. 887389

$10\sqrt{2}\,kg$

$2\sqrt{10}\,kg$

10 kg

20 kg

139 A block of mass 5 kg is kept inside a train accelerating at $2 m/s^2$ rightward. The pseudo force acting on the block is: 887388

2.5 N leftward

10 N rightward

10 N leftward

50 N rightward

140 **Identify correct statement about pseudo force.** 887387

It is observed in inertial frame and acts opposite to the direction acceleration of the frame.

It is observed in non inertial frame and acts in the direction of acceleration of the frame.

It is observed in non inertial frame and acts in the opposite direction of acceleration of the frame.

It is observed in non inertial frame and acts in the opposite direction of velocity of the frame.

141 **Two springs of spring constants 100 N/m and 300 N/m are connected in series. The equivalent spring constant is:** 887386

50 N/m

75 N/m

100 N/m

400 N/m

142 **The spring constant k has the same SI unit as:** 887385

Pressure

Force per unit length

Energy per unit volume

Momentum

143 **A particle moves 5 m east and then 5 m north. The ratio of distance to displacement is:** 887384

$\sqrt{2}:1$

$\sqrt{3}:\sqrt{2}$

$\sqrt{2}:\sqrt{5}$

$\sqrt{2}:\sqrt{3}$

144 **If the momentum of a body is doubled, its kinetic energy increases by:** 887383

300%

400%

200%

40%

145 **Which of the following quantities is always a scalar?** 887382

Force

Momentum

Work

Displacement

146 The work done by the gravitational force on a body moving in a closed path is: 887381

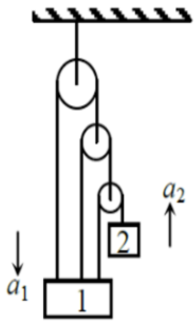
Positive

Negative

Zero

Infinite

147 887380



Find the relation between a1 and a2 :

$a_1 = 3a_2$

$a_2 = 3a_1$

$a_2 = 6.a_1$

$a_2 = 7.a_1$

148 **A block of mass m is attached to the lower end of a vertical spring. Initially, the spring is unstretched, and the block is held at rest. If the block is released from this position, what will be the maximum extension produced in the spring?** 887379

2mg/k

mg/k

3mg/2k

mg/2k

149 **A force of 30 N acts on a body for 6 s. If the mass of the body is 4 kg and it starts from initial velocity 15 m/s, its final kinetic energy is** 887378

4050 J

3600 J

7200J

14400 J

150 **The ratio of maximum height to horizontal range for a projectile projected at 45 degree from horizontal is** 887377

1 : 2

1 : 4

1 : 1

2 : 1

151 The equation of the trajectory of a projectile is $y = \frac{x}{\sqrt{3}} - \frac{x^2}{60}$ **887376**

The horizontal range of the projectile is:

20√3 m

30√3 m

40√3 m

60√3 m

152 A 4 kg body has a gravitational potential energy of 240 J. If $g=10m/s^2$, its height above the reference level is **887375**

4 m

6 m

8 m

10 m

153 A spring stores 32 J when compressed by 40 cm. The force required to maintain this compression is **887374**

80 N

120 N

160 N

200 N

154 **A block of mass 2 kg is pulled by a horizontal force of 30 N on a μ rough surface (= 0.2). If it moves 5 m from rest, the increase in its kinetic energy is (Take g=10 m/s^2)** 887373

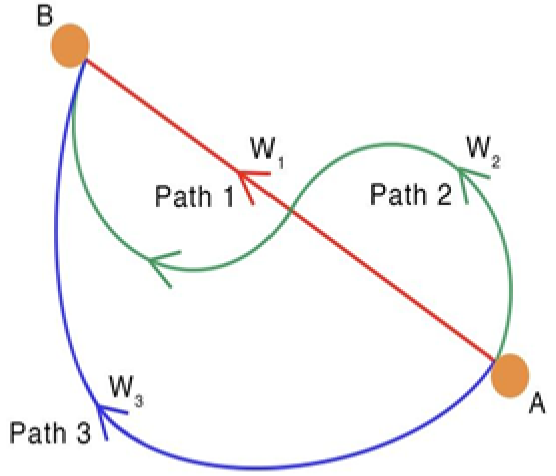
110 J

130 J

150 J

170 J

155 **A body travels from point A to B along three different paths as shown. The work done by conservative force in three cases will be** 887372



$W_1 = W_2 = W_3$

$W_3 > W_2 = W_1$

$W_1 \neq W_2 \neq W_3$

$W_1 > W_2 = W_3$

156 **Which of the following forces can perform non-zero work over a closed path?** 887371

Gravitational force

Electrostatic force

Spring force

Frictional force

157 **Find the quantity which is not a scalar.** 887370

work

Speed

energy

momentum

158 **A shell of mass 200 g is fired by a gun of mass 100 kg. If the muzzle speed of the shell is 80 m/s, calculate the recoil speed of the gun** 887369

16 cm/s

8m/s

4 cm/s

16m/s

159 $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$ represents a unit vector when c is 887368

0.2

$\sqrt{0.2}$

$\sqrt{0.8}$

0

160 The work done by a force $\vec{F} = \left(-6x^3\hat{i}\right)\text{N}$, in displacing a particle from x=4m to x=-2m is 887367

360 J

240 J

– 240 J

– 360 J

161 **The average resisting force that must act on a 5 kg mass to reduce its speed from 65 cm/s to 15 cm/s in 0.2s is** **887366**

12.5 N

25 N

50 N

100 N

162 **A net force of 15 N acts on a 3 kg body moving initially at 4 m/s. If the force acts through 8 m, the final kinetic energy is:** **887365**

24 J

96 J

144 J

168 J

163 **A body of mass 2 kg is lifted vertically through 5 m. Take g=10m/s^2. Increase in potential energy is:** **887364**

50 J

100 J

150 J

200 J

164 Which of the following always does negative work? 887363

Gravity on a freely falling body

Friction on a moving block

Applied force in the direction of motion

Tension while lifting a body upward

165 An object of 1 kg mass has a momentum of 10 kg m/sec then the kinetic energy of the object will be 887362

100 J

50 J

1000 J

200 J

166 The potential energy of a particle moving in a three-dimensional space is given by $U(x,y,z)=2x^2y$ The force acting on the particle is: 887361

$4xy\hat{\imath} + 2x^2\hat{\jmath}$

$-4xy\hat{\imath} - 2x^2\hat{\jmath}$

$-2x^2\hat{\imath} - 4xy\hat{\jmath}$

$4xy\hat{\imath} - 2x^2\hat{\jmath}$

167 **A body of mass 2 kg is thrown up vertically with K.E. of 490 joules. If the acceleration due to gravity is 9.8 m/s^2 , then the height at which the K.E. of the body becomes half its original value is given by** 887360

50 m

12.5 m

25 m

10 m

168 **Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched by 10cm and 40cm respectively. They will have potential energy in the ratio** 887359

1 : 32

1 : 16

1 : 8

1 : 64

169 A force $\vec{F} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ acts on a particle and produces a displacement of $\vec{s} = 2\hat{i} - 3\hat{j} + x\hat{k}$. If the work done is zero, the value of x is

887358

– 2

½

6

2

170 A position dependent force *F=7-2x+3x²* newton acts on a small body of mass 2 kg and displaces it from x=0 to x=5m. The work done in joules is

887357

70

270

35

135

171 Which one of the following is not a conservative force

886070

Gravitational force

Electrostatic force between two charges

Magnetic force between two magnetic dipoles

Frictional force

172 **The IUPAC name of the compound having the formula $CH \equiv C-CH = CH_2$ is** **810596**

1-butyne-3-ene

but-1-yne-3-ene

1-butene-3-yne

3-butene-1-yne.

173 **The shape of PF_5 molecule is:** **806396**

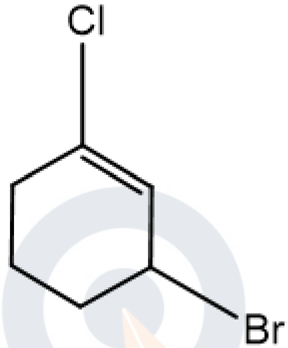
Trigonal bipyramidal

Tetrahedral

Octahedral

Square pyramidal

174 **The IUPAC name of compound shown below is** **803872**



2-bromo-6- chlorocyclohex-1-ene

6-bromo-2-chlorocyclohexene

3-bromo-1-chlorocyclohexene

1-bromo-3-chlorocyclohexene

175 The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is 803477

$-COOH, -SO_3H, -CONH_2, -CHO$

$-SO_3H, -COOH, -CONH_2, -CHO$

$-CHO, -COOH, -SO_3H, -CONH_2$

$-CONH_2, -CHO, -SO_3H_1' - COOH$

176 Which of the following molecular orbitals has two nodal planes 801065

$\sigma 2s$

$\pi 2p_y$

$\pi^* 2p_y$

$\sigma^* 2p_x$

177 The potential energy of a body is given by, $U = A\text{-}Bx^2$ (Where x is the displacement). The magnitude of force acting on the particle is 790667

Constant

Proportional to x

Proportional to x^2

Inversely proportional to x

178 × A spring of spring constant 5 10 ^{3} N/m is stretched initially by 5cm from the unstretched position. Then the work required to stretch it further by another 5cm is

790665

6.25 N-m

12.50 N-m

18.75 N-m

25.00 N-m

179 A particle of mass 0.3 kg is subjected to a force F=-kx with k=15 N/m. What will be its initial acceleration if it is released from a point 20 cm away from the origin

790059

5 m/s^2

10 m/s^2

3 m/s^2

15 m/s^2

180 A particle of mass m at rest is acted upon by a force F for a time t. Its Kinetic energy after an interval t is 775896

$\frac{F^2 t^2}{m}$

$\frac{F^2 t^2}{2m}$

$\frac{F^2 t^2}{3m}$

$\frac{F\,t}{2m}$